

04/22/2026

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Compact generators for  
spectral presheaves

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§ the FLTz Stratification

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Last time Finished the proof that for  $(N, \Sigma)$  projective, the lax sym mon functor

$$R_{\Sigma} \lim_{\sigma \in \Sigma^{\text{op}}} \text{Mod}_{\omega_{\sigma v}} \text{Sh}(M_R) \longrightarrow \text{Sh}(M_R)$$

is an equivalence. Showed that for any cone  $\sigma \in N_R$ , the functor

$$\Psi_{\sigma} : \text{Fun}(\Theta(\sigma)^{\text{op}}, \text{Sp}) \longrightarrow \text{Mod}_{\omega_{\sigma v}} \text{Sh}(M_R)$$

is fully faithful when restricted to the image of the spectral Yoneda embedding.

- Realized that in order to see that  $\Psi_{\sigma}$  is fully faithful, it suffices to find a compactly generated subcategory

$$D_{\sigma} \subset \text{Sh}(M_R)$$

Closed under colimits such that  $\Psi_{\sigma}$  factors through  $D_{\sigma}^{\omega}$  and  $\Psi_{\sigma}$  is fully faithful on generators.

Today Compact generators for Spectral presheaves and identifying  
such a  $D_0$ .

## Compact generators for Spectral presheaves

**Exercise** Let  $\mathcal{C}$  be a stable  $\infty$ -category and let  $f: X \rightarrow Z$  and  $g: Y \rightarrow Z$  be maps. There is a natural identification

$$X \underset{Z}{\times} Y \simeq \text{fib}(f - g: X \oplus Y \rightarrow Z)$$

**Cor** Let  $\mathcal{C}$  be a stable  $\infty$ -category and let  $\mathcal{C}_0 \subset \mathcal{C}$  be a full subcategory closed under finite colimits and shifts. Then  $\mathcal{C}_0$  is closed under finite limits.

**Proof**

By assumption,  $0 \in \mathcal{C}_0$  and  $\mathcal{C}_0$  is closed under finite direct sums. So by the exercise, it suffices to show that  $\mathcal{C}_0$  is closed under fibers. Recall that

$$\text{fib}(f) \simeq \text{cofib}(f)[-1]$$

So the claim follows from the closure of  $\mathcal{C}_0$  under finite colimits and shifts. □

**Lem** Let  $\mathcal{C}$  be an  $\infty$ -category with filtered colimits. Then the full subcategory  $\mathcal{C}^w \subset \mathcal{C}$  is closed under finite colimits and retracts. If  $\mathcal{C}$  is also stable, then  $\mathcal{C}^w \subset \mathcal{C}$  is also closed under finite limits.

**Proof**

Closure under retracts is clear from the fact that the class of equivalences is closed under retracts.

For finite colimits, let  $X_\bullet: I \rightarrow \mathcal{C}^w$  be a finite diagram and let  $Y_\bullet: J \rightarrow \mathcal{C}$  be a filtered diagram. Then since filtered colimits and finite limits of anima commute, we see that

$$\mathrm{Map}_{\mathcal{C}}\left(\mathrm{colim}_{i \in I} X_i, \mathrm{colim}_{j \in J} Y_j\right) \simeq \lim_{i \in I^{\mathrm{op}}} \mathrm{Map}_{\mathcal{C}}\left(X_i, \mathrm{colim}_{j \in J} Y_j\right)$$

$$\simeq \lim_{i \in I^{\mathrm{op}}} \mathrm{colim}_{j \in J} \mathrm{Map}_{\mathcal{C}}(X_i, Y_j)$$

$$\simeq \mathrm{colim}_{j \in J} \lim_{i \in I^{\mathrm{op}}} \mathrm{Map}_{\mathcal{C}}(X_i, Y_j)$$

$$\simeq \operatorname{Colim}_{j \in J} \operatorname{Map}_{\mathcal{C}} \left( \operatorname{Colim}_{i \in I} X_i, Y_j \right)$$

Thus  $\operatorname{Colim}_{i \in I} X_i$  is compact.

Now assume that  $\mathcal{C}$  is stable. By the previous Corollary, it suffices to show that  $\mathcal{C}^{\omega} \subset \mathcal{C}$  is closed under shifts. This follows immediately from the definition of a compact object and the fact that for each  $n \in \mathbb{Z}$ , the shift functor

$$(-)[n]: \mathcal{C} \rightarrow \mathcal{C}$$

is an equivalence. □

**Prop.** Let  $\mathcal{C}_0$  be a small  $\infty$ -category. Then  $\operatorname{Fun}(\mathcal{C}_0, \operatorname{Sp})$  is compactly generated. Moreover,  $\operatorname{Fun}(\mathcal{C}_0, \operatorname{Sp})^{\omega}$  is the smallest full subcategory of  $\operatorname{Fun}(\mathcal{C}_0, \operatorname{Sp})$  that contains the image of the spectral Yoneda embedding and is closed under finite colimits, retracts, and shifts.

> This isn't difficult, but we won't prove it here.

⚠ It is tempting to think that  $\text{Fun}(\mathcal{C}_0^{\text{op}}, \mathcal{S}\mathcal{P})^{\omega} = \text{Fun}(\mathcal{C}_0^{\text{op}}, \mathcal{S}\mathcal{P}^{\omega})$ .  
 This is true when  $\mathcal{C}_0$  is a finite poset, but false in general. We'll revisit this point later.

Lem. Let  $G: \mathcal{A} \rightarrow \mathcal{B}$  be an exact functor between stable  $\infty$ -categories, and write

$$\ker(G) = \{X \in \mathcal{A} \mid G(X) = 0\}$$

Then  $\ker(G)$  is closed under finite limits, finite colimits, and retracts.

Proof. Clear from the definitions. □

Lemma. Let  $F: \mathcal{C} \rightarrow \mathcal{D}$  be an exact functor between stable  $\infty$ -categories. Write  $\mathcal{C}' \subset \mathcal{C}^{\text{op}} \times \mathcal{C}$  for the full subcategory spanned by those  $(X, Y) \in \mathcal{C}^{\text{op}} \times \mathcal{C}$  such that the natural map

$$\phi_{X,Y}: \underline{\text{Map}}_{\mathcal{C}}(X, Y) \longrightarrow \underline{\text{Map}}_{\mathcal{D}}(F(X), F(Y))$$

is an equivalence. Then:

↑ mapping spectra

(1) For all  $X, Y \in \mathcal{C}$ , we have  $(X, 0), (0, Y) \in \mathcal{C}'$

(2) The full subcategory  $\mathcal{C}' \subset \mathcal{C}^{\text{op}} \times \mathcal{C}$  is closed under finite limits, finite colimits, and retracts

Proof.

(1) is obvious.

For (2), consider the functor given by the composite

$$G: \mathcal{C}^{\text{op}} \times \mathcal{C} \longrightarrow \text{Fun}([1], \text{Sp}) \xrightarrow{\text{fib}} \text{Sp}$$

$$(X, Y) \longmapsto [\phi_{X,Y}: \underline{\text{Map}}_{\mathcal{C}}(X, Y) \longrightarrow \underline{\text{Map}}_{\mathcal{D}}(F(X), F(Y))]$$

then take the fiber

Since mapping spectra are exact in each variable, taking fibers is exact, and  $F$  is exact, we see that  $G$  is exact.

Moreover,  $\phi_{X,Y}$  is an equivalence if and only if  $\text{fib}(\phi_{X,Y}) = 0$ .

So  $\mathcal{C}' = \text{Ker}(G)$ . Thus the claim follows from the previous result.  $\square$



Cor Let  $\mathcal{C}_0$  be a small  $\infty$ -category, let  $D$  be a stable  $\infty$ -cat with colimits, and let  $F: \text{Fun}(\mathcal{C}_0^{\text{op}}, \mathcal{S}\mathcal{P}) \rightarrow D$  be a colimit-preserving functor. Then:

(1) If the restriction of  $F$  to the image of the spectral Yoneda embedding factors through  $D^{\omega}$ , then the restriction of  $F$  to  $\text{Fun}(\mathcal{C}_0^{\text{op}}, \mathcal{S}\mathcal{P})^{\omega}$  factors through  $D^{\omega}$ .

(2) If for all  $X$  and  $Y$  in the image of the spectral Yoneda embedding,

$$\underline{\text{Map}}_{\mathcal{C}}(X, Y) \xrightarrow{\sim} \underline{\text{Map}}_D(F(X), F(Y)),$$

then the same is true for  $X, Y \in \text{Fun}(\mathcal{C}_0^{\text{op}}, \mathcal{S}\mathcal{P})^{\omega}$ . In particular, the restriction of  $F$  to  $\text{Fun}(\mathcal{C}_0^{\text{op}}, \mathcal{S}\mathcal{P})^{\omega}$  is fully faithful.

(3) If (1) and (2) are satisfied, then  $G$  is fully faithful.

Upshot. From what we've shown so far, to see that  $\Psi_\sigma$  is fully faithful, it suffices to find a compactly generated subcat  $D_\sigma \subset \text{Sh}(M_R)$  closed under colimits such that for each  $z \in \text{Ob}(\sigma)$ , we have  $\omega_z \in D_\sigma^\omega$ .

## Constructible Sheaves

- > We'll consider a generalization of locally constant sheaves that includes extension by zeros of constant sheaves

Def. Let  $(P, \leq)$  be a poset. The Alexandroff topology on the set  $P$  is the topology where

$U \subseteq P$  is open iff  $p \in U$  and  $q > p$  implies  $q \in U$ .

- > So for each  $p \in P$ , the subsets

$$P_{\geq p} = \{q \in P \mid q \geq p\}$$

and

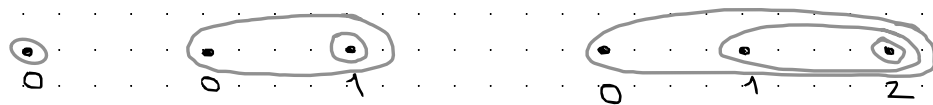
$$P_{> p} := \{q \in P \mid q > p\}$$

are open.

- > Similarly,  $P_{\leq p}$  and  $P_{< p}$  are closed.

Note Each point  $\{p\} = P_{\geq p} \cap P_{\leq p}$  is locally closed

Ex The topologies on  $[n] = \{0 < \dots < n\}$ :



Def Let  $P$  be a poset. A  $P$ -stratification of a space  $X$  is a continuous map  $s: X \rightarrow P$

> For each  $p \in P$ , the locally closed subspace  $X_p = s^{-1}(p)$  is called the  $p$ th stratum of  $X$ .

If  $s: X \rightarrow P$  and  $t: Y \rightarrow Q$  are stratified spaces, a map of stratified spaces  $(X, P) \rightarrow (Y, Q)$  consists of a map of posets  $\phi: P \rightarrow Q$  and a continuous map  $f: X \rightarrow Y$  making the square

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ s \downarrow & & \downarrow t \\ P & \xrightarrow{\phi} & Q \end{array}$$

commute

Ex.  $\left\{ \begin{array}{l} \text{stratifications} \\ \text{of } X \text{ by } [n] \end{array} \right\} \simeq \left\{ \begin{array}{l} \text{filtrations by closed} \\ \text{subspaces} \\ F_0(X) \subset F_1(X) \subset \dots \subset F_n(X) = X \end{array} \right\}$

$$s: X \rightarrow [n] \longmapsto F_i(X) = s^{-1}(\{0 < \dots < i\})$$

$$[x \mapsto \min\{i \mid x \in F_i(X)\}] \longleftarrow F_\bullet(X)$$

Def. Let  $(X, P)$  be a stratified space and  $\mathcal{E}$  a presentable  $\infty$ -category. Write  $i_p: X_p \hookrightarrow X$  for the inclusion of the  $p$ th stratum. A sheaf  $F \in \text{Sh}(X; \mathcal{E})$  is  $P$ -constructible if for each  $p \in P$ , the sheaf  $i_p^*(F)$  is locally constant. We write

$$\text{Cons}_P(X, \mathcal{E}) \subset \text{Sh}(X; \mathcal{E})$$

for the full subcategory spanned by the constructible sheaves.

Lemma Let  $P$  be a poset. Then  $(P_{\geq p})_{p \in P}$  forms a basis for the topology on  $P$ . As a result,

$$\text{Sh}^{\text{hyp}}(P, \mathcal{E}) \xrightarrow{\sim} \text{Fun}(P, \mathcal{E})$$

If  $P$  satisfies the ascending chain condition, then every sheaf is a hypersheaf.

## The FLTZ Stratification

> In order to understand what's going on, let's consider the cases of  $A^1$  and  $A^2$

**Example** Consider the lattice  $\mathbb{Z}$  with cone  $\sigma = \mathbb{R}_{\geq 0}$ , so that  $X_\sigma \simeq A^1$ . In this case,  $\sigma^\vee = \mathbb{R}_{\geq 0}$  as well, and

$$\Theta(\sigma) \simeq \{m + \mathbb{R}_{\geq 0} \mid m \in \mathbb{Z}\} \simeq \mathbb{Z}$$

In this case, we have

$$\omega_{m+\sigma^\vee} \simeq \omega_{\mathbb{R}_{>m}} \simeq \mathbb{1}_{\mathbb{R}_{>m}} [1]$$

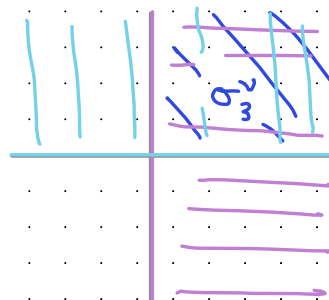
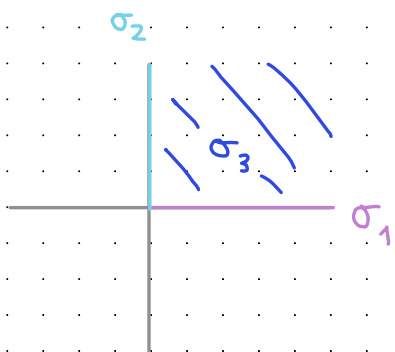
In particular, each  $\omega_{m+\sigma^\vee}$  is constant on  $\mathbb{R}_{>m}$  with value 1, and zero on  $\mathbb{R}_{\leq m}$ . Hence each  $\omega_{m+\sigma^\vee}$  is constructible for the stratification of  $\mathbb{R}$  by  $\mathbb{R} \setminus \mathbb{Z}$  and  $\mathbb{Z}$ .

Ex. Consider the lattice  $\mathbb{Z}^2$  with fan generated by  $e_1$  and  $e_2$ . So there are three cones

$$\sigma_1 = \mathbb{R}_{\geq 0} e_1, \quad \sigma_2 = \mathbb{R}_{\geq 0} e_2, \quad \sigma_3 = \text{Cone}(e_1, e_2)$$

With duals

$$\sigma_1^\vee = \text{right half plane}, \quad \sigma_2^\vee = \text{upper half plane}, \quad \sigma_3^\vee = \text{Cone}(e_1^*, e_2^*)$$

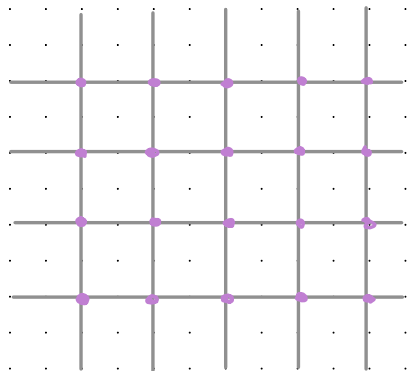


Observe that for each  $\sigma \in \{\sigma_1, \sigma_2, \sigma_3\}$ , and  $m \in \mathbb{Z}^2$ , the sheaf

$$\omega_{m+\sigma v} \cong \omega_{m+\sigma v_{\sigma 0}} \cong \mathbb{1}_{m+\sigma v_{\sigma 0}}[2]$$



is constructible for the stratification given by the following affine hyperplane arrangement.



The strata are  $\mathbb{Z}^2$ , the grey line segments, and the interiors of the remaining squares.

> We're interested in a generalization to arbitrary fans.

Recall, For a cone  $\sigma \in N_{\mathbb{R}}$ , we write

$$\sigma^{\perp} := \{ v \in M_{\mathbb{R}} \mid \text{for all } u \in \sigma \text{ we have } \langle u, v \rangle = 0 \}.$$

If  $\sigma$  is generated by a single element,  $\sigma^{\perp}$  is a hyperplane.

Ntn. For a fan  $(N, \Sigma)$ , write  $\Sigma(1)$  for the set of 1-dimensional cones in  $\Sigma$ .

Construction (FLTZ Stratification) Let  $(N, \Sigma)$  be a fan, and assume that  $\Sigma$  spans  $N_{\mathbb{R}}$ . We define a stratification  $M_{\mathbb{R}} \rightarrow P_{\Sigma}$  by a poset of Krull dimension  $n = \dim(M_{\mathbb{R}})$  as follows. Consider the affine hyperplane arrangement

$$H_{\Sigma} := \{m + \sigma^{\perp} \mid m \in M, \sigma \in \Sigma(1)\}$$

Define a filtration  $\text{Fil}_{\bullet}^{\Sigma} M_{\mathbb{R}}$  on  $M_{\mathbb{R}}$  as follows.

- >  $\text{Fil}_n^{\Sigma}(M_{\mathbb{R}}) = M_{\mathbb{R}}$
- >  $\text{Fil}_{n-1}^{\Sigma}(M_{\mathbb{R}}) = \bigcup_{h \in H_{\Sigma}} h$
- > In general,  $v \in M_{\mathbb{R}}$  is in  $\text{Fil}_{n-i}^{\Sigma}(M_{\mathbb{R}})$  if and only if  $v$  belongs to at least  $i$  distinct hyperplanes in  $H_{\Sigma}$ .

This gives a stratification over  $\{0 < \dots < n\}$ . We further

refine the stratification by declaring that each connected component of  $(M_{\mathbb{R}})_k$  is its own stratum. Then the set  $P_{\Sigma}$  of strata becomes a poset under the relation

$$S \leq T \text{ if and only if } S \subset \overline{T}.$$

We refer to this stratification as the FLTZ stratification.

Observe. The FLTZ stratification only depends on the set of 1-cones  $\Sigma(1) \subset \Sigma$ .

**Lemma.** Let  $(N, \Sigma)$  be a fan such that  $\Sigma$  spans  $N_{\mathbb{R}}$ . Then for each  $\sigma \in \Sigma$ , and  $z \in \Theta(\sigma)$ , the sheaf  $\omega_z \in \text{Sh}(M_{\mathbb{R}})$  is constructible with respect to the FLTZ stratification.

**Proof.**

There is an  $m \in M$  such that  $z = m + \sigma^{\vee}$ . Moreover,

$$\omega_z \cong \mathbb{1}_{m + \sigma^{\vee}, 0}[\dim(M_{\mathbb{R}})],$$

so it suffices to see that  $\mathbb{1}_{m + \sigma^{\vee}, 0}$  is constructible.

With respect to the FLTZ stratification. This is immediate from the definition of the FLTZ stratification and the fact that  $\mathbb{1}_{m+\sigma^{\vee,0}}$  is constant on  $m+\sigma^{\vee,0}$  and zero outside of it.

□